FIITJEE

ALL INDIA TEST SERIES

OPEN TEST

JEE (Main)-2019

TEST DATE: 10-02-2019

Time Allotted: 3 Hours **Maximum Marks: 360**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only one section: **Section-A**.

  Section-A (01 – 30, 31 – 60, 61 – 90) *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

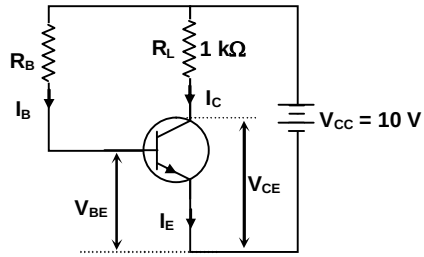
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. In the circuit shown in the figure the current gain, $\beta = 100$ for the transistor. Find the bias resistance R_B so that $V_{CE} = 6$ V.
(Neglect V_{BE})

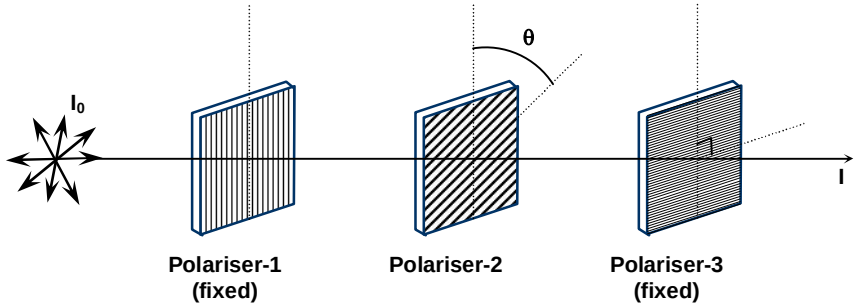


(A) $8 \times 10^5\ \Omega$
(B) $4 \times 10^5\ \Omega$
(C) $6 \times 10^5\ \Omega$
(D) $2.5 \times 10^5\ \Omega$

2. In a nuclear reactor, moderators slow down the neutrons which come out in a fission process. The moderators used have light nuclei. Heavy nuclei will not solve the purpose because
(A) The net weight of the reactor would be unbearably high.
(B) Heavy nuclei will break generating unwanted smaller nuclei.
(C) Elastic collision of neutrons with heavy nuclei will not slow them down.
(D) Substances with heavy nuclei do not occur in liquid or gaseous state at room temperature.

3. Identify the INCORRECT reasoning about why an audio signal of 10 KHz frequency cannot be transmitted over long distances without modulation.
(A) The minimum size of antenna required is at least 7.5 km and this is not convenient.
(B) Effective power transmitted would be very low if the size of antenna is less than 7.5 km.
(C) Audio signals cannot be transmitted through sky waves.
(D) Audio signals can be transmitted only to a small distance by space wave.

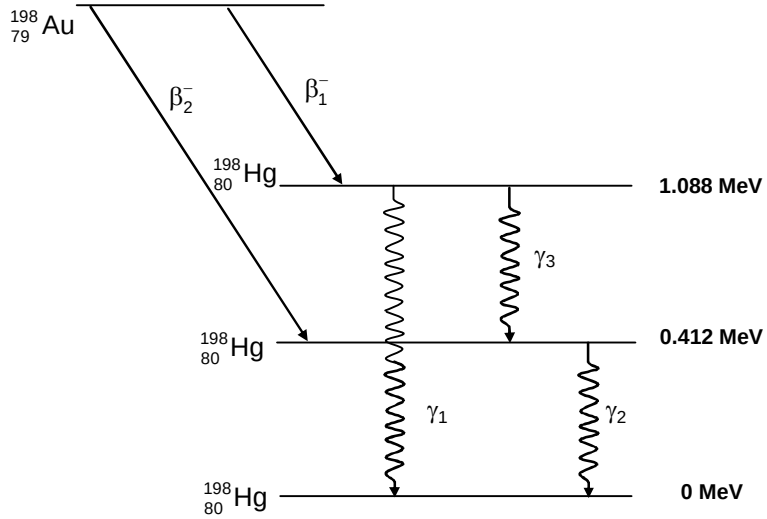
4. A student writes down four conclusions that he observes while performing a polarization experiment with 3 polarisers as shown in the figure. The leftmost and the rightmost polariser are fixed and crossed and the middle one can be rotated. The intensity of the unpolarised light is I_0. Identity the INCORRECT conclusion noted by him.



(A) If $\theta = 0°$ then $I = 0$
(B) If $\theta = 90°$ then $I = 0$
(C) If $\theta = 45°$ then $I = \frac{I_0}{8}$
(D) If $\theta = 30°$ then $I = \frac{3I_0}{16}$

5. The focal lengths of the objective and eyepiece of an astronomical telescope are 4 m and 2 cm respectively. The diameter of the objective is 50 cm. The telescope is directed to view two distant stars (almost point sources of light). The sources are roughly at the same distance ($\approx 10^5$ light year) along the line of sight but are separated transverse to the line of sight by 10^{11} m. Now answer the correct statement when these stars are observed through this telescope.
   (A) The images of these stars will become clear if the diameter of the objective lens is increased.
   (B) The image of these stars will become clear if the diameter of the eyepiece is increased.
   (C) Limit of resolution of this telescope is $\approx 10^{-10}$ rad.
   (D) Two stars cannot be resolved by this telescope.

6. In the decay process of a nucleus given below, identify the correct statement.
[given: m(Au^{198}) = 197.968233 u, m(Hg^{198}) = 197.966760 u]

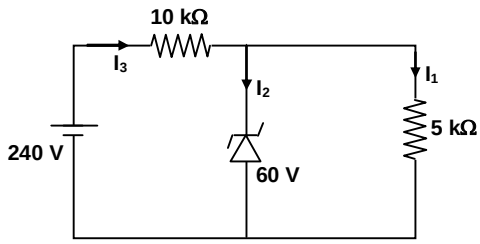


   (A) Maximum K.E. of $\beta_1^- = 0.960$ MeV
   (B) Maximum K.E. of $\beta_2^- = 0.284$ MeV
   (C) Energy associated with γ_1 is 0.412 MeV
   (D) Energy associated with γ_3 is 0.676 MeV

7. The simple Bohr model cannot be directly applied to calculate the energy levels of an atoms with many electrons. This because
   (A) of screening effects
   (B) the force between the nucleus and an electron will no longer be given by Coulomb's law.
   (C) of the electrons not being subject to central force
   (D) of the electrons colliding with each other.

8. The main scale division and pitch of a screw gauge is 0.05 cm. Number of circular scale division in the circular scale is 100. When no object is placed between the studs then the '0' of main scale is not visible and 5th division of circular scale coincides with the main scale. When a wire is placed in between the studs then the reading on main scale is 4.6 mm and 52nd division on circular scale matches with reference line on main scale. The actual diameter of wire is
   (A) 4.385 mm
   (B) 4.885 mm
   (C) 5.335 mm
   (D) 4.835 mm

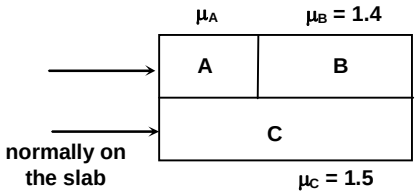
9. Find the correct value of current (i_1, i_2 and i_3) in the circuit shown in the figure.



(A) I_1 = 16 mA, I_2 = 8 mA, I_3 = 8 mA
(B) I_1 = 16 mA, I_2 = 0 mA, I_3 = 16 mA
(C) I_1 = 12 mA, I_2 = 6 mA, I_3 = 18 mA
(D) I_1 = 12 mA, I_2 = 4 mA, I_3 = 16 Ma

10. In a series LCR circuit the voltage across the resistance, capacitor and inductor is 20 V each. If the inductor is short circuited then, the voltage across the capacitor is
(A) 20 V
(B) $10\sqrt{2}$ V
(C) $20\sqrt{2}$ V
(D) $20(\sqrt{2}-1)$V

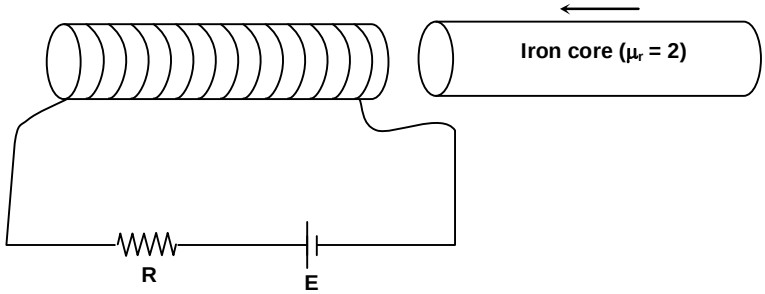
11. A composite slab is made of three transparent materials as shown in the figure. Monochromatic parallel beam of light is incident normally on the slabs A and C. Thickness of B is twice that the thickness of A. The number of waves in C is equal to the number of waves in the combination of A and B. If the refractive index of material of B is μ_B = 1.4 and that of C is μ_C = 1.5. Then,



(A) The refractive index of A is 1.6
(B) The refractive index of A is 1.7
(C) The refractive index of A is 1.55
(D) Frequency of light in A is 2 times the frequency of light in B

12. A monochromatic beam of light of wavelength λ = 4500 Å is incident normally on the slits of young's double slit set up. The separation between the slits is 9 μm and the slit width is 2 μm for both the slits. The distance between the slits and the screen is 1 m. Identify the correct option assuming that intensity of the incident light is sufficiently high.
(A) 8 interference fringes will be formed in the central diffraction fringe.
(B) 9 diffraction fringes will be formed in the central interference fringe.
(C) 9th interference maximum will not be formed.
(D) The 2nd diffraction maximum will not be formed.

13. A metallic core made of paramagnetic substance (μ_r =2) and of same size as that of the air core solenoid (self induction L) is inserted suddenly (t = 0) within the space of the core solenoid as shown in the figure. (ignore resistance of the solenoid as well as the end effects) and assume solenoid is of infinite length. Identify the correct option. Assume that charge flowing to the circuit is negligible.



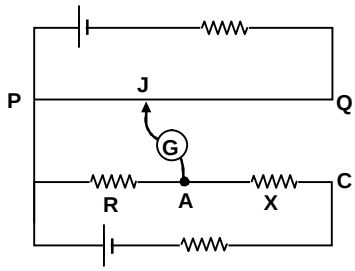
(A) Current just after introduction of the metallic cone is E/R.

(B) Current just before inserting the metallic cone and long time after the introduction of metallic core are not same.

(C) Current at time t after inserting the iron core is $\frac{E}{R}\left(1-2e^{-\frac{Rt}{L}}\right)$

(D) Current at time t after inserting the iron cone is $\frac{E}{2R}\left(2-e^{-\frac{Rt}{2L}}\right)$

14. The circuit for measurement of resistance by potentiometer is as shown in the figure. The galvanometer is first connected at point A and zero deflection is observed at length PJ = 20 cm. In second case galvanometer is connected to the point C and the zero deflection is observed at a length 50 cm from P. Then the unknown resistance X is



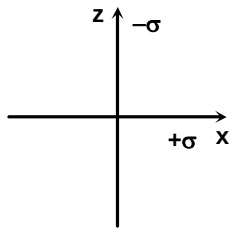
(A) $\frac{5R}{2}$

(B) 2R

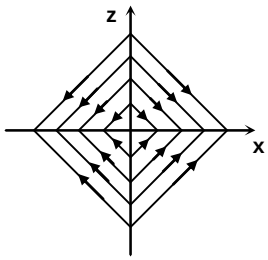
(C) $\frac{2R}{5}$

(D) $\frac{3R}{2}$

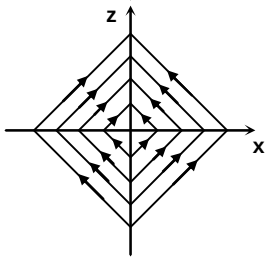
15. Two infinitely large charged non-conducting planes having uniform surface charge density $+\sigma$ and $-\sigma$ are placed along x-y and y-z plane respectively as shown in the figure. Then the nature of the electric lines of forces in x-z plane is best represented by



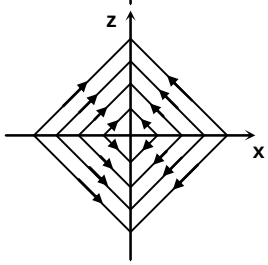
(A)

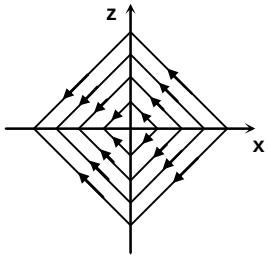


(B)

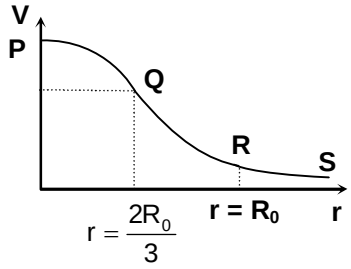


(C)

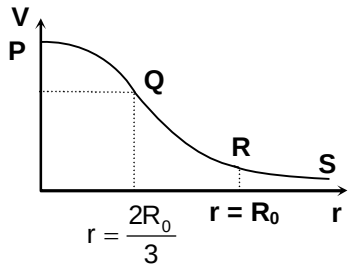


(D) 

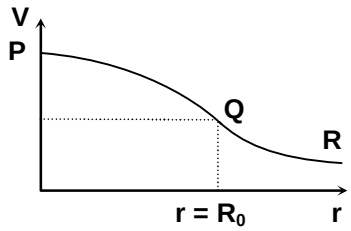
16. A non-conducting solid sphere has a volumetric charge density $\rho = \rho_0\left(1-\frac{r}{R_0}\right)$, where ρ_0 is constant, R_0 is the radius of sphere and 'r' is the distance from the centre of sphere. The graph between potential and distance from the centre of sphere is best represented by

(A) 

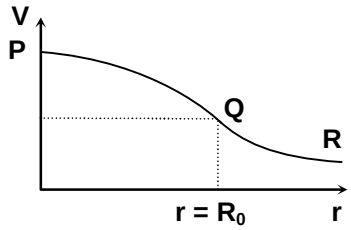
PQ curve is parabolic
QR curve is parabolic
RS curve is rectangular hyperbola

(B) 

PQ curve is cubic
QR curve is cubic
RS curve is rectangular hyperbola

(C) 

PQ curve is cubic
QR curve is rectangular hyperbola

(D) 

PQ curve is parabola
QR curve is rectangular hyperbola

17. A particle is moving in x-y plane with $\vec{v} = v_x\hat{i} + v_y\hat{j}$ and acceleration $\vec{a} = a_x\hat{i} + a_y\hat{j}$. If the magnitude of velocity of the particle is constant then choose the correct option.

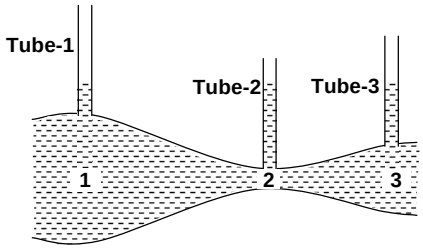
(A) $a_x v_x + a_y v_y = 0$

(B) $a_x v_x - a_y v_y = 0$

(C) $\frac{a_x}{v_x}+\frac{a_y}{v_y}=0$

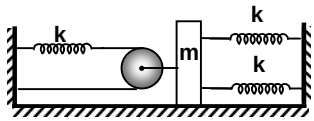
(D) $\frac{a_x}{v_x}-\frac{a_y}{v_y}=0$

18. An ideal liquid is flowing in a tube of small variable cross-section. Tube –1, tube- 2, and tube –3 is connected at point 1, 2 and 3 respectively as shown. Let h_1, h_2 and h_3 are the final heights of liquid in tube 1, 2 and 3 respectively and v_1, v_2 and v_3 are the velocity of liquid at cross-section 1, 2 and 3 respectively.



(A) $h_1 > h_3$ and $v_1 > v_3$
(B) $h_2 < h_3$ and $v_2 < v_3$
(C) $h_1 < h_3$ and $v_1 > v_3$
(D) $h_1 > h_3$ and $v_1 < v_3$

19. A block of mass m is connected with three massless springs and a massless pulley, springs were in natural length initially. Surface is smooth and block is slightly shifted towards right and released then its time period is



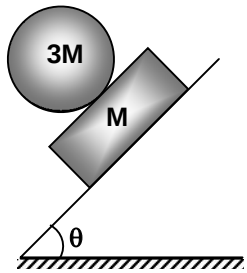
(A) $2\pi\sqrt{\frac{m}{6k}}+2\pi\sqrt{\frac{m}{2k}}$

(B) $2\pi\sqrt{\frac{m}{6k}}+\pi\sqrt{\frac{m}{2k}}$

(C) $\pi\sqrt{\frac{m}{6k}}+\pi\sqrt{\frac{m}{2k}}$

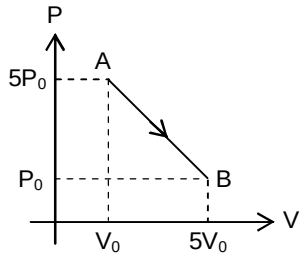
(D) $\pi\sqrt{\frac{m}{6k}}+2\pi\sqrt{\frac{m}{2k}}$

20. A plank of mass M and sphere of mass 3 M are kept on a smooth inclined plane as shown in the figure. The coefficient of friction between sphere and plank is 'μ'. If system is released from rest then choose the correct alternative at t = 0.



(A) Friction force between sphere and plank is $\mu Mg\cos\theta$.
(B) Acceleration of plank with respect to sphere is non zero.
(C) Friction force between sphere and plank is zero.
(D) Friction force between sphere and plank is $3\mu mg\cos\theta$.

21. One mole of an ideal diatomic gas undergoes a process AB as shown. Pressure at A and B are $5P_0$ and P_0 while volumes are V_0 and $5\ V_0$ respectively. During the process from $A \rightarrow B$, pressure and volume of the gas, where the direction of flow of heat is reversed, is



(A) $\frac{5}{2}P_0,\frac{7}{2}V_0$

(B) $P_0, 5V_0$

(C) $4P_0, 2V_0$

(D) $\frac{9}{2}P_0, \frac{3}{2}V_0$

22. A disc of radius R = 2m starts rolling in vertical plane on a rough horizontal surface. It is observed that the magnitude of acceleration of a point inside the sphere at a distance $\sqrt{2}$ m from the centre is zero. Then find the vertical height of point from the ground.

(A) $\frac{1}{4}$ m

(B) $\frac{3}{2}$ m

(C) $\frac{3}{4}$m

(D) 1 m

23. A equation is given by $C = \frac{Q^2 \cdot \Delta\theta}{m \cdot \gamma}$, where C is specific heat, $\Delta\theta$ is difference in temperature, Q is heat given to the system and m is mass of system. Then the dimension of γ is given by

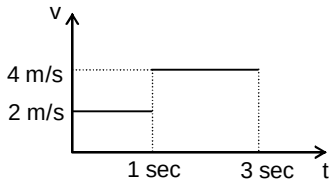
(A) $ML^2T^{-2}\theta^2$

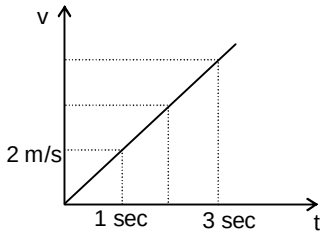
(B) $ML^2T^{-2}\theta^{-2}$

(C) $M^2L^2T^2\theta^2$

(D) $ML^2T^{-1}\theta^{-1}$

24. Two particles P and Q starts from the origin along x-axis. Velocity time graph of both particles are shown in the figure. During the given time interval, the maximum separation between the particles is





(A) 4 m

(B) 1 m

(C) 2 m

(D) 3 m

25. A lamp post is fixed at point (0, 0, 5m). A man is moving with constant velocity 6 m/s along a line $y = \sqrt{3}x$. Height of man is 2m and initially the man is at origin. The velocity of edge of his shadow along the line of motion of man is

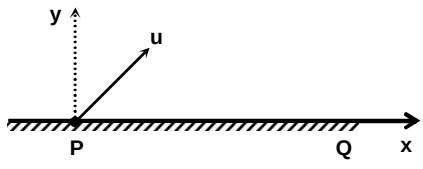
(A) $8\hat{i} + 6\hat{j}$

(B) $6\hat{i} + 8\hat{j}$

(C) $5\sqrt{3}\,\hat{i} + 5\hat{j}$

(D) $5\hat{i} + 5\sqrt{3}\hat{j}$

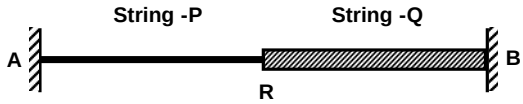
26. A small particle is projected from point 'P' in a vertical plane with an initial speed u. It hits the ground at Q. Assume that its collision with the ground which occurs at Q is perfectly inelastic. If area covered between the trajectory and x-axis by particle from P to Q is maximum, then the angle of projection of the particle is



(A) 60°

(B) 45°
(C) 30°
(D) 75°

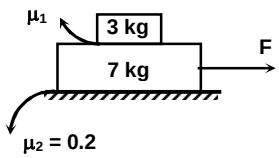
27. Two strings 'P' and 'Q' of length 1 meter and 1/2 meter respectively are connected at 'A' and 'B'. 'R' is the junction of both strings. The mass per unit length of strings P and Q are 0.16 kg/m and 0.25 kg/m respectively. A tension of 100 N is maintained in both strings. A pulse of amplitude 2 mm is given at point 'A' it partly reflected and partially transmitted at R. Then the amplitude of reflected and transmitted wave is



(A) $\frac{16}{9}$ mm and $\frac{1}{9}$ mm
(B) $\frac{8}{9}$ mm and $\frac{2}{9}$ mm
(C) $\frac{2}{9}$ mm and $\frac{16}{9}$ mm
(D) $\frac{2}{9}$ mm and $\frac{8}{9}$ mm

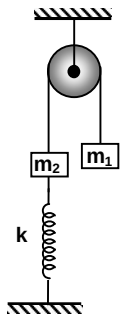
28. Over a certain region of space. The gravitational potential is given by $v = 3x^2y - 2yz + x^3z$. The magnitude of gravitational field at (0, 1, –1) is
(A) 4
(B) 2
(C) $2\sqrt{2}$
(D) 1

29. Two blocks of mass 3 kg and 7 kg are kept on a rough horizontal surface. A force F is applied on lower block of mass 7 kg. Coefficient of friction between 3 kg and 7 kg is 'μ_1' and between 7 kg and surface it is $\mu_2 = 0.2$. Choose the correct option.



(A) If applied force is 15 N then slipping will occur between 3 kg and 7 kg.
(B) If $\mu_1 = 0$ and applied force is 30 N then both block have same acceleration of magnitude 3 m/s^2.
(C) Slipping will start between 7 kg and surface for minimum value of F = 20 N and it is independent of μ_1.
(D) For F = 10 N, slipping can be possible between 7 kg and 3 kg it will depend on μ_1.

30. Two blocks of mass $m_1 = 3$ kg and $m_2 = 1$ kg are connected through a spring of spring constant K = 100 N/m in a vertical plane. Pulley is massless and frictionless. Initially the spring is at its natural length. Now, if system is released in vertical plane then maximum elongation in the spring is
(A) 1/5 m
(B) 3/5 m
(C) 2/5 m
(D) 6/5 m



Chemistry

PART – II

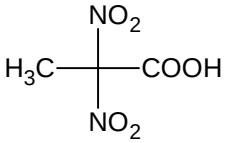
SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

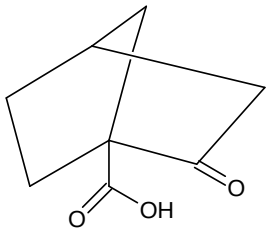
31. If x is number of σ bonds, y be the number of π bonds and z be the number of bonding electrons respectively in C_2 molecule, then the value of $\frac{z-x}{y}$ is:

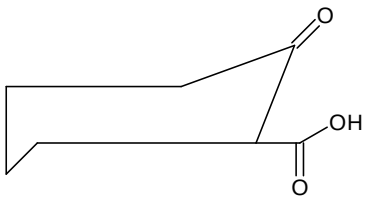
(A) 7
(B) ∞
(C) 4
(D) 0

32. Which of the following acid is least likely to decarboxylate simply on heating?

(A) 

(B) $H_2C{=}CH{-}COOH$

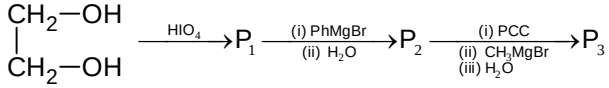
(C) 

(D) 

33. The species having the strongest gas phase proton affinity amongst the following is:

(A) N^{3-}
(B) NF_3
(C) NH_3
(D) $N(CH_3)_3$

34. Which of the following is **incorrect** for the final product of the given sequence.



(A) Compound P_3 on oxidation with (CrO_3 + H_2SO_4) gives a compound which gives 2, 4 DNP Test.
(B) Compound P_3 on reaction with I_2 + NaOH gives yellow precipitate .
(C) Compound P_3 on reaction with ceric ammonium nitrate gives red colouration
(D) Compound P_3 on reaction with MnO_2 gives carboxylic acid

35. What is the magnetic moment of iron in Brown coloured ring complex formed in nitrate test?
(A) Zero
(B) $\sqrt{15}$
(C) $\sqrt{2}$
(D) $\sqrt{35}$

36. Calculate the value of E_{cell} for the given cell:

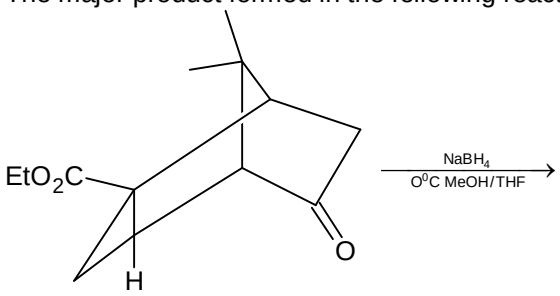
$$Pt\,|\,H_2(g)\,|\,\underset{(0.1M)}{CH_3COOH} + \underset{(0.1M)}{CH_3COONa}\,||\,\underset{(Ksp=4\times10^{-12})}{Ag_2S}\,|\,Ag(s)$$

$\frac{2.303RT}{nF} = 0.059$, $Ka(CH_3COOH) = 10^{-5}$, $E^0_{Ag^+/Ag} = 0.8V$, $\log 5 = 0.698$

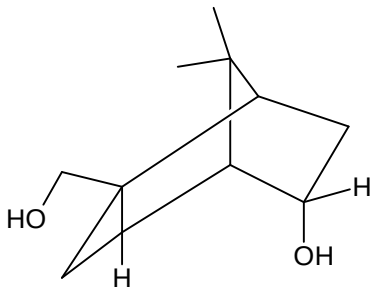
(A) 0.8590 V
(B) 0.6230 V
(C) 0.8767 V
(D) 0.6407 V

37. For a reaction 2A + B $\rightarrow$ 3Z, if the rate of consumption of A is 2×10^{-4} mol/dm^3/s, the rate of formation of Z (in mole^{-1}s^{-1}):
(A) 3×10^{-4}
(B) 2×10^{-4}
(C) $\frac{4}{3} \times 10^{-4}$
(D) 4×10^{-4}

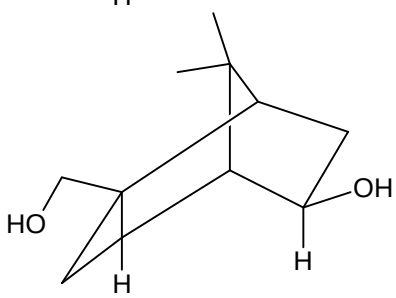
38. The major product formed in the following reaction is:



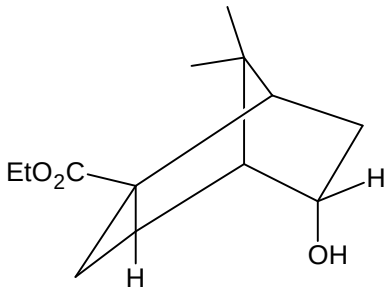
(A)



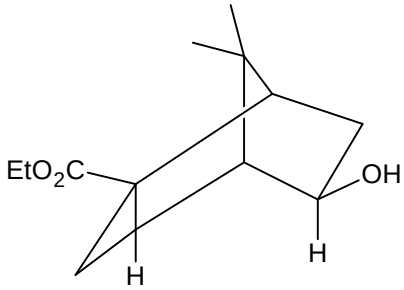
(B)



(C)



(D)



39. Which is not the property of diamond?
(A) each carbon atom is sp^3 hybridized
(B) it is an isomer of graphite
(C) it is purest form of carbon
(D) It has abnormally high m.p.

40. For a homogeneous reaction $A \rightarrow 3B$, if the pressure after time 't' was P_T and after completion it was P_∞, then choose the correct relationship:

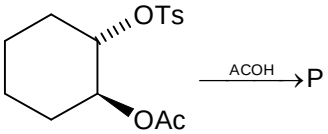
(A) $K = \frac{1}{t}\ln\left(\frac{P_\infty}{3P_\infty - 3P_T}\right)$

(B) $K = \frac{1}{t}\ln\left(\frac{2P_\infty}{P_\infty - P_T}\right)$

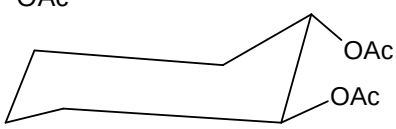
(C) $K = \frac{1}{t}\ln\left(\frac{3P_\infty}{2P_\infty - P_T}\right)$

(D) $K = \frac{1}{t}\ln\left(\frac{2P_\infty}{3P_\infty - 3P_T}\right)$

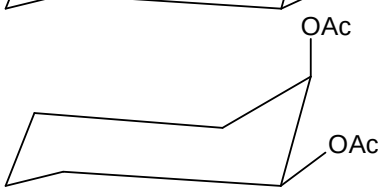
41. Major product of the following reaction in its most stable form is



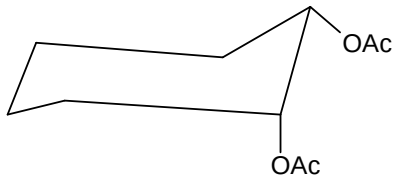
(A)



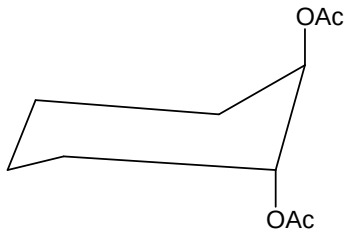
(B)



(C)

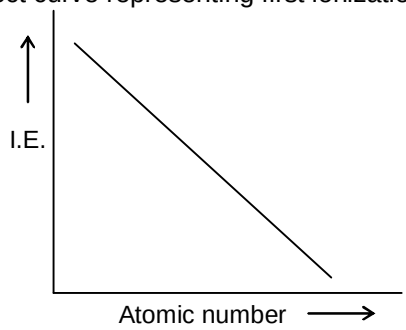


(D)

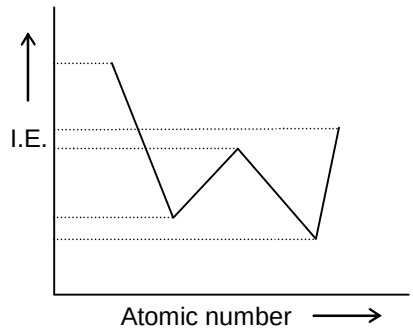


42. Hydrogen gap is found in which block of the periodic table?
(A) s-block
(B) p-block
(C) d-block
(D) f-block

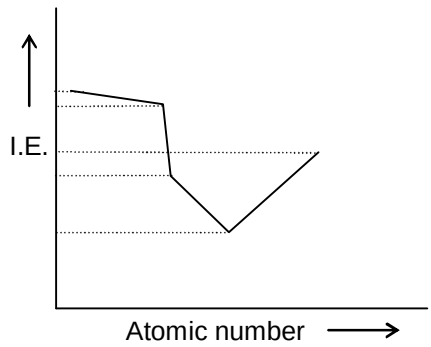
43. The correct curve representing first ionization energy of group 13 elements is:
(A)



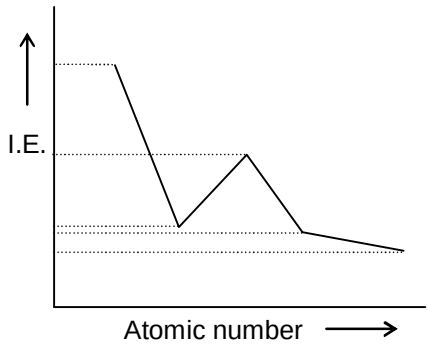
(B)



(C)



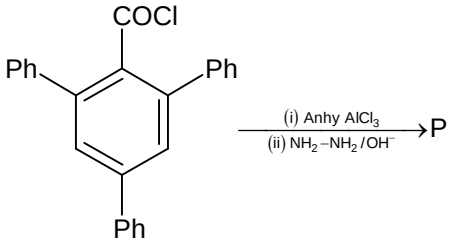
(D)



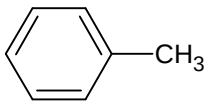
44. Among the following complexes (1) $[Co(ox)_3]^{-3}$, (2) trans-$[CoCl_2(en)_2]^+$, (3) $[Cr(EDTA)]^-$, the chiral one(s) is/are:
(A) 1 and 2
(B) 3 and 2
(C) 3 only
(D) 1 and 3

45. Which statement(s) concerning Bohr's model is **incorrect**?
(A) Predicts the probability of electron near nucleus is more
(B) Angular momentum of electron in H-atom = $\frac{nh}{2\pi}$
(C) Introduces the idea of stationary states.
(D) Explain line spectrum of hydrogen atom.

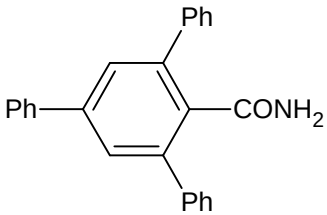
46. The final product in the given sequence is:



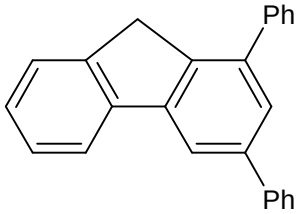
(A)



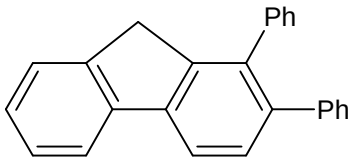
(B)



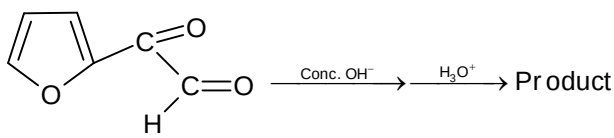
(C)



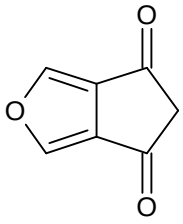
(D)



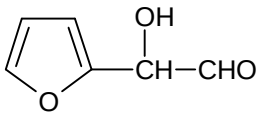
47. The final product of the following sequence is:



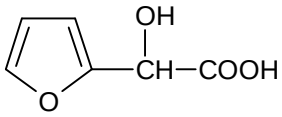
(A)



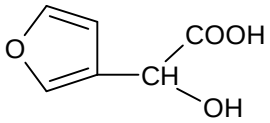
(B)



(C)



(D)



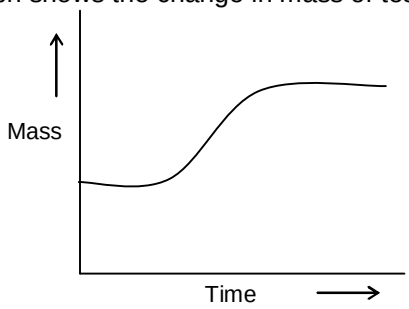
48. What is the number of chloride in the given list which dissolves in excess of NaOH solution?
$SnCl_2$, $CrCl_3$, $AlCl_3$, $FeCl_3$, $ZnCl_3$, $BeCl_2$, AgCl
(A) 7
(B) 5
(C) 3
(D) 4

49. Potassium chlorate, $KClO_3$ is heated in test tube and following reaction occurs at 25°C.

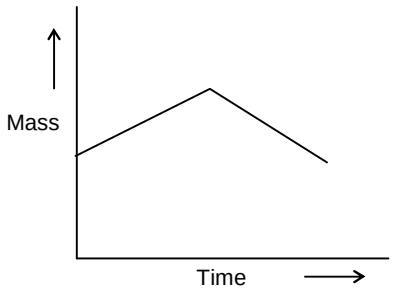
$$KClO_3 \xrightarrow{\Delta} KCl + \frac{3}{2}O_2$$

Which graph shows the change in mass of test tube and its content?

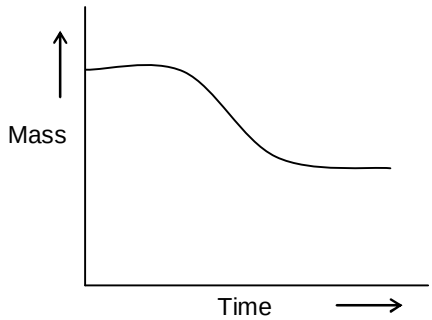
(A)



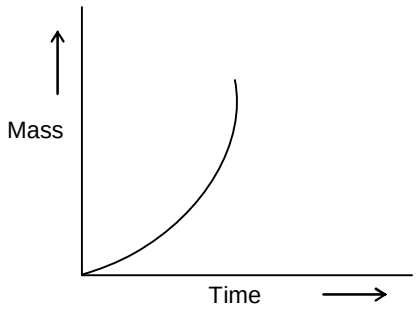
(B)



(C)



(D)



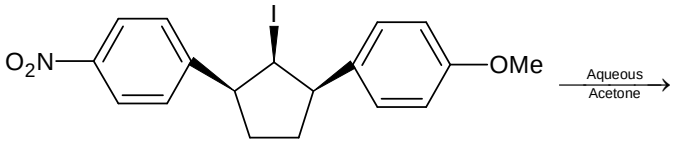
50. Graph between $\log\frac{x}{m}$ vs log P is straight line inclined at an angle of 45°. When the pressure is 0.5 atm and log K is 0.699, the amount of solution adsorbed per g of adsorbent will be:
(A) 2.5 g/g adsorbent
(B) 0.25 g/g adsorbent
(C) 1 g/g adsorbent
(D) 1.5 g/g adsorbent

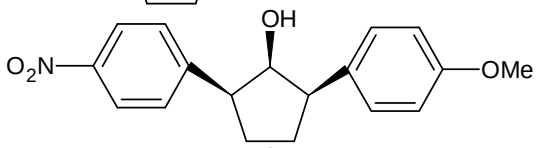
51. An organic compound is reacted with HNO_2 at 0°C and then the resulting solution is added to a solution of β-napthol whereby a brilliant red dye is produced. This indicates the presence of:
(A) $–NO_2$ group
(B) $–CONH_2$ group
(C) Aromatic NH_2 group
(D) Aliphatic NH_2 group

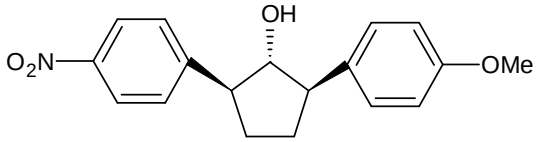
52. 'Matte' in the extraction of Cu has the composition:
(A) Cu_2O, Cu_2S, FeS
(B) Cu_2O, FeO, Cu_2S
(C) Cu_2O, FeO
(D) Cu_2S, FeO, FeS

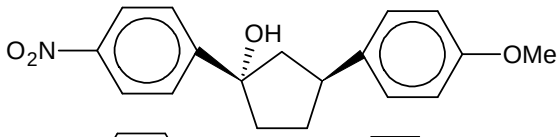
53. Select an **incorrect** statement out of the following:
(A) Instead of Radical substitution, cyclopropane undergoes electrophilic addition reactions in sun light.
(B) In general, bromination is more selective than chlorination
(C) The 2,4,6-tri-tert butylphenoxy radical is resistant to dimerization
(D) The boiling point and dipole moment of R – X (R = CH_3) is greater when X = F than when X = Cl.

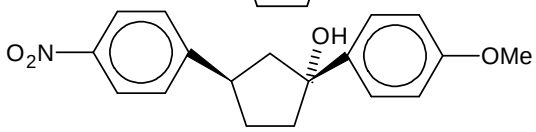
54. Identify the major product for the following reaction:



(A)


(B)


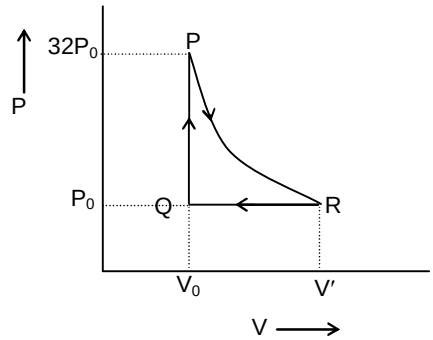
(C)


(D)


55. The pair that yields the same gaseous product on reaction with water is:
(A) Ca and CaH_2
(B) Ba and BaO_2
(C) KH and KO_2
(D) Na and Na_2O_2

56. A crystal is constituted of X and Y particles. Y form ccp arrangement where as X occupies all octahedral voids. If all the particles along one body diagonal are removed and Z atom are introduced in the available voids of that body diagonal (lying inside the cube completely), the formula of this solid becomes $X_aY_bZ_c$. The value of a, b and c respectively are:
(A) 3, 4, 3
(B) 4, 5, 4
(C) 16, 15, 12
(D) 3, 15, 4

57. One mole of a monoatomic ideal gas is taken along the given path as shown in the P–V diagram. $Q \rightarrow P \rightarrow R \rightarrow Q$



The process $P \rightarrow R$ is purely isothermal, then calculate the work done for the process $R \rightarrow Q$.

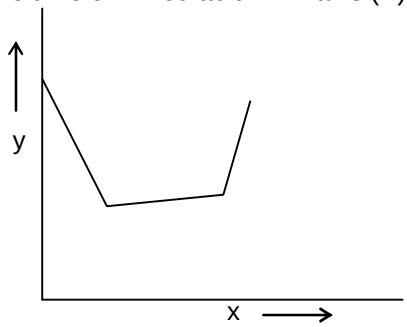
(A) $24P_oV_o$
(B) $36P_oV_o$
(C) $31P_oV_o$
(D) P_oV_o

58. If N and S both are present in an organic compound during Lassaigne's test, both will change into

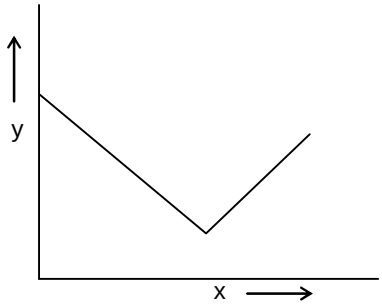
(A) Na_2S and NaCN
(B) NaSCN
(C) Na_2SO_3 and NaCN
(D) Na_2S and NaCNO

59. On titrating a NaOH solution conductometrically with a mixture of HCl and CH_3COOH solution, plot of the volume of mixed acid in Y–axis (Y) against conductance in X-axis (X) will look like:

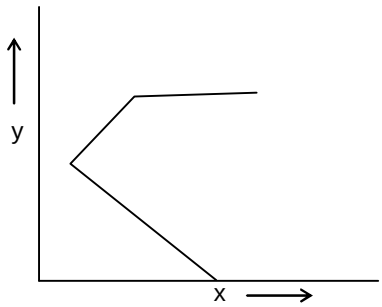
(A)



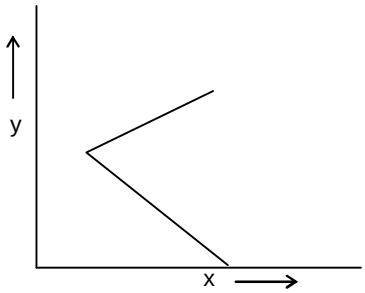
(B)



(C)



(D)



60. How many isomeric disubstituted borazine are possible?
(A) 6
(B) 3
(C) 5
(D) 4

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. An equilateral triangle is inscribed in ellipse whose equation is $x^2 + 4y^2 = 4$, one vertex of triangle is (0, 1) and one altitude is contained in y-axis and length of each side is $\sqrt{\frac{m}{n}}$ (where m and n are relatively prime), then m + n is
(A) 937
(B) 973
(C) 793
(D) 739

62. If $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{4}{r^2+r+16}\right) = \tan^{-1}\left(\frac{\alpha}{10}\right)$, then α is
(A) 80
(B) 40
(C) 20
(D) 30

63. The radius of the circle touching the parabola $y^2 = 4x$ at an end of latus rectum and passing through focus is
(A) $\sqrt{5}$
(B) $\sqrt{2}$
(C) $\sqrt{3}$
(D) $\sqrt{7}$

64. If $9 + f''(x) + f'(x) = x^2 + f^2(x)$, f(x) is a twice differentiable function such that $f''(x) \neq 0 \ \forall \ x \in R$ and let P be the point of maxima of f(x), then number of tangents which can be drawn from P to circle $x^2 + y^2 = 9$
(A) 0
(B) 2
(C) 1
(D) nothing can be said

65. The number of five digit numbers that can be formed using all the digits 0, 1, 3, 6, 8 greater than 30,000 that are divisible by 11 is
(A) 10
(B) 8
(C) 12
(D) 16

66. The vertices of a triangle are 3 + 4i, 4 + 3i and $2\sqrt{6} + i$, then distance between orthocentre and circumcentre of the triangle is equal to
(A) $\sqrt{137 - 28\sqrt{6}}$
(B) $\sqrt{137 + 28\sqrt{6}}$

(C) $\frac{1}{2}\sqrt{137+28\sqrt{6}}$

(D) $\frac{1}{3}\sqrt{137+28\sqrt{6}}$

67. If $\int \frac{1-7\cos^2 x}{\sin^7 x \cdot \cos^2 x}dx = \frac{g(x)}{\sin^7 x}+c$, then $g'(0)+g''\left(\frac{\pi}{4}\right)$ is

(A) 1
(B) 17
(C) 9
(D) 5

68. The value of $\int_0^2\left(\sqrt{x^3+1}\right)dx+\int_0^2\sqrt[3]{x^2+2x}\,dx$ is

(A) 6
(B) 4
(C) 8
(D) 2

69. Three distinct numbers are chosen one by one without replacement from set {1, 2, 3, ..., 9}. If it is known that sum of first two chosen numbers is even, then the probability that sum of all three numbers is even is

(A) $\frac{15}{28}$

(B) $\frac{11}{28}$

(C) $\frac{13}{28}$

(D) $\frac{17}{28}$

70. If curve $y = k\cos\frac{x}{2}$ divides the area bounded by y = sin x, x-axis in between x = 0 and x = π in two equal parts, then k is

(A) $2-\sqrt{2}$
(B) $2+\sqrt{2}$
(C) $3-\sqrt{2}$
(D) $3+\sqrt{2}$

71. If A and B are non-singular square matrices of even order such that $A^T = A, B^T = -B$ and AB + BA = O (where O is a null matrix), then

(A) (AB) is skew symmetric matrix
(B) A^{-1} and B^{-1} commute each other
(C) $(A^{-1}B)$ is symmetric matrix
(D) B and A^{-1} commute each other

72. If x and y are independent variables and $f(x) = 1 + \int_0^1 (x^2y^2 + x) f(y)\,dy$ be a continuous and differentiable function, then f′(0) is

(A) $\frac{164}{57}$

(B) $\frac{25}{19}$

(C) $\frac{38}{73}$

(D) $\frac{91}{103}$

73. If $\vec{a}, \vec{b}, \vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angle $\alpha \in \left(0, \frac{\pi}{2}\right)$ with each of $\vec{a}$, $\vec{b}$ and $\vec{c}$, then $\left[\left|\vec{a} + \vec{b} + \vec{c} + \vec{d}\right|^2\right]$ is equal to (where [.] denotes the greatest integer function)

(A) 5
(B) 6
(C) 7
(D) 8

74. Let g : [1, 6] → [0, ∞) be a real values differentiable function satisfying $g'(x) = \frac{2}{x + g(x)}$ and g(1) = 0, then the maximum value of g cannot exceed

(A) $\ell n 2$

(B) $\ell n 6$

(C) $3\ell n 2$

(D) $2\ell n 6$

75. Number of solutions of the equation $\frac{\sqrt{6} - \sqrt{2}}{\sin x} + \frac{\sqrt{2} + \sqrt{6}}{\cos x} = 8$ in $\left(0, \frac{\pi}{2}\right)$ is/are

(A) 1
(B) 0
(C) 2
(D) 4

76. Let $T = \int_0^{\ell n 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} dx$, then e^T is

(A) $\frac{7}{4}$

(B) $\frac{11}{8}$

(C) $\frac{11}{2}$

(D) $\frac{11}{4}$

77. Let F(x) be a non-negative continuous function defined on R such that $F(x)+F\left(x+\frac{1}{2}\right)=3$, then $\int_0^{1500} F(x)dx$ is equal to

(A) 2250
(B) 1500
(C) 750
(D) 0

78. The value of $\lim_{n\to\infty}\left(\frac{1}{2n+1}+\frac{1}{2n+3}+\frac{1}{2n+5}+.....+\frac{1}{4n-1}\right)$ (where $n \in N$) is

(A) $\ell n\ 2$
(B) $\ell n\ \sqrt{2}$
(C) $2\ \ell n\ 2$
(D) $3\ \ell n\ \sqrt{2}$

79. A particle starts to travel from a point P on the curve C_1: $|z - 3 - 4i| = 5$ where $|z|$ is maximum. From P, the particle moves through an angle $\tan^{-1}\frac{3}{4}$ in anticlockwise direction on $|z - 3 - 4i| = 5$ about the centre and reaches at point Q. From Q, it comes down parallel to imaginary axis by 2 units and reaches at point R. Complex number corresponding to point R in the Argand plane is

(A) (3 + 5i)
(B) (3 + 7i)
(C) (3 + 8i)
(D) (3 + 9i)

80. The parabola $x = y^2 + ay + b$ intersect the parabola $x^2 = y$ at (1, 1) at right angle, then the area enclosed by the parabola $x = y^2 + ay + b$ and its latus rectum is

(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) $\frac{1}{4}$
(D) $\frac{1}{6}$

81. A plane P is perpendicular to the vector $\vec{A}=2\hat{i}+3\hat{j}+6\hat{k}$ and contains the terminal point of the vector $\vec{B}=\hat{i}+5\hat{j}+3\hat{k}$. The perpendicular distance from the origin to the plane P is equal to

(A) 2 units
(B) 5 units
(C) 6 units
(D) 7 units

82. Find the number of all 6 digit numbers such that all the digits of each number are from the set {1, 2, 3, 4, 5} and any digit that appears in the number appears at least twice, is

(A) 1205
(B) 1305
(C) 1405
(D) 1505

83. There are two set of observation with term $\{x_r, f_r\}$ and $\{y_r, f_r\}$ (where x_r and y_r are observation with their respective frequency f_r). Now if $x_r = 2^r$ and $y_r = 2^r - 2$ and $f_r = {}^nC_r$. If the sets are having their variance V_A and V_B respectively, then $\frac{V_A + V_B}{V_B}$ is equal to

(A) 3
(B) 2
(C) –2
(D) 1

84. The preposition $[\sim(p \to q) \to (q \to p)] \Leftrightarrow [\sim\{(p \to q) \wedge (\sim q \wedge p)\}]$ is a
(A) Tautology
(B) Fallacy
(C) Contingency
(D) none of these

85. If $\int_0^1 \ell nx \cdot \ell n(1-x)dx = p - \frac{\pi^q}{r}$, p, q, r $\in$ N, then p + q + r is equal to $\left[\text{Use } \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{r^2} = \frac{\pi^2}{6}\right]$
(A) 10
(B) 9
(C) 5
(D) 8

86. Two flagstaffs stand on a horizontal plane. A and B are two points on the line joining their feet and between them. The angles of elevation of the tops of the flagstaffs as seen from A are 30º and 60º and as seen from B are 60º and 45º. If AB is 30 m, the distance between the flagstaffs in metres is
(A) $30+15\sqrt{3}$
(B) $45+15\sqrt{3}$
(C) $60-15\sqrt{3}$
(D) $60+15\sqrt{3}$

87. The line x + y + 2 = 0 is a tangent at point A, intersect the directrix at B and tangent at vertex at C respectively. If the focus of parabola is S(2, 0), then value of AC·BC is
(A) 8
(B) 16
(C) 4
(D) 10

88. Two variable chords AB and BC of a circle $x^2 + y^2 = r^2$ are such that AB = BC = r. The locus of the point of intersection of tangents at A and C is
(A) $x^2 + y^2 + 3r^2$
(B) $x^2 + y^2 = 4r^2$
(C) $x^2 + y^2 = 5r^2$
(D) $x^2 + y^2 = 6r^2$

89. If $\frac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right).....\left((2n-1)^4+\frac{1}{4}\right)}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right).....\left((2n)^4+\frac{1}{4}\right)} = \frac{1}{k_1n^2+k_2n+k_3}$, then $k_1 - k_2 + k_3$ is equal to

(A) 8
(B) 5
(C) 3
(D) 1

90. If the value of $\lim_{n\to\infty} n\tan(4\pi en!) = L$, then fundamental period of f(x) = |sin x| + |cos x| is

(A) $\frac{L}{12}$
(B) $\frac{L}{8}$
(C) $\frac{L}{4}$
(D) $\frac{L}{6}$